

# ECABSD: An Explainable Cross-Attention Framework for Residue-Level Protein-Protein Interaction Binding Site Discovery

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**Background:** Many biological processes depend on PPIs. Current models predict binding sites in isolation, neglecting cross-attention between target and partner proteins. ECABSD is introduced as an interaction-aware framework integrating structural and sequential features.

**Methods:** ECABSD uses bidirectional cross-attention, GATv2 structural encoding, and ESM-2 embeddings. A hybrid Focal + Soft Dice loss with dynamic thresholding addresses class imbalance. Grad-CAM and Attention Rollout provide visual explanations.

**Results:** Random split: F1=0.7010, ROC-AUC=0.9373. Homology-filtered ( $\leq 30\%$ ): F1=0.5797, ROC-AUC=0.8928 — outperforming all published baselines on MCC and ROC-AUC. Ablation confirms cross-attention is most critical ( $-0.169$  F1). Case study 1AY7: Precision=0.938, Recall=1.000, F1=0.968.

**Conclusion:** ECABSD's attention-based integration proves effective for PPI site prediction with interpretable insights for experimental validation.

**Keywords:** Protein-Protein Interaction, Binding Site Discovery, Deep Learning, GATv2, ESM-2, Cross-Attention, Explainable AI

## 1. Introduction

Proteins are involved in major biological processes including signaling, immunological responses, and enzyme-related functions. Protein-protein interactions (PPIs) are essential for protein activity modulation. When PPIs fail, dysfunctions associated with neurodegenerative diseases and cancer can occur. Experimental methods such as yeast two-hybrid screening and X-ray crystallography are valuable but expensive and labor-intensive, driving the need for computational approaches.

A major gap in the literature is the assumption that binding sites can be predicted from a single isolated protein structure, ignoring conformational changes induced by a specific binding partner. Most datasets are severely imbalanced — binding sites make up only a small fraction of the protein surface. Deep neural networks often behave as black boxes, limiting biological trust.

ECABSD addresses these challenges by using structural and sequence information from both the target protein (Chain A) and the partner protein (Chain B). GATv2 captures residue-level interactions; ESM-2 embeddings capture evolutionary patterns; cross-attention enables bidirectional contextual learning for biologically relevant

interface representation.

## 2. Related Work

Early predictors such as SPPIDER, ProMate, and PSIVER relied on sequence conservation and accessible surface area with SVMs, achieving F1 scores of 0.45–0.50. PAIRpred introduced pairwise residue scoring; DELPHI added deep learning features (F1  $\approx$  0.55). MaSIF-site uses geometric deep learning on molecular surfaces (F1  $\approx$  0.60) but predicts from single isolated chains. ECABSD is the first method combining GATv2 structural encoding with partner-aware cross-attention for per-residue PPI binding site prediction.

## 3. Method

### 3.1 Architecture

Input PDB structures for Chain A and Chain B are processed through: (1) graph construction with ESM-2 embeddings as 33-dimensional node features at a 10.0 Å C $\alpha$ -C $\alpha$  cutoff, (2) six GATv2 layers (256-dim, 4 heads, residual connections, GELU), (3) global context pooling added to local residue features, (4) bidirectional cross-attention fusion (Chain A as Q, Chain B as K/V), and (5) a 3-layer MLP prediction head with LayerNorm, ReLU, Dropout(0.3), and Sigmoid.

### 3.2 Loss and Training

The hybrid loss  $L = 0.6 \times L_{\text{Focal}} + 0.4 \times L_{\text{Dice}}$  combines Focal Loss ( $\alpha=0.90$ ,  $\gamma=2.0$ ) for class imbalance with Soft Dice Loss for F1-correlated overlap. Training uses AdamW ( $\text{lr}=3\times10^{-4}$ ), Cosine Annealing with 15-epoch Linear Warmup, gradient clipping at 1.0, and dynamic PR thresholding.

### 3.3 Datasets

ECABSD is trained on 3,816 protein-protein complexes (PDBbind + DIPS subset), evaluated on Docking Benchmark 5 (DB5). MMseqs2 clustering at  $\leq 30\%$  sequence identity eliminates homology leakage. Chain-swap augmentation ( $p=0.50$ ) doubles training data. Interfacial residue labels use a 4.5 Å atomic contact cutoff.

## 4. Results and Discussion

### 4.1 Main Results

Table 1: ECABSD V3 Performance Metrics (113,112 test residues)

| Metric          | Random Split  | Homology-Filtered ( $\leq 30\%$ ) | Description                        |
|-----------------|---------------|-----------------------------------|------------------------------------|
| Accuracy        | 0.8989        | 0.8828                            | Overall correct predictions        |
| Precision       | 0.6396        | 0.5305                            | Predicted binding sites correct    |
| Recall          | 0.7756        | 0.6389                            | Actual binding sites detected      |
| <b>F1-Score</b> | <b>0.7010</b> | <b>0.5797</b>                     | <b>Harmonic mean P/R</b>           |
| <b>MCC</b>      | <b>0.6452</b> | <b>0.5152</b>                     | <b>Imbalanced-data correlation</b> |
| <b>AUC-ROC</b>  | <b>0.9373</b> | <b>0.8928</b>                     | <b>Area under ROC curve</b>        |
| AUC-PR          | 0.7462        | 0.6077                            | Area under PR curve                |

## 4.2 Baseline Comparison

Table 2: Baseline Comparison (Homology-Filtered Evaluation)

| Method                  | F1            | MCC           | ROC-AUC       | Reference        |
|-------------------------|---------------|---------------|---------------|------------------|
| SPPIDER                 | 0.48          | 0.25          | n/a           | Porollo 2007     |
| ProMate                 | 0.45          | 0.22          | n/a           | Neuvirth 2004    |
| PSIVER                  | 0.47          | 0.24          | n/a           | Murakami 2010    |
| PAIRpred                | 0.52          | 0.30          | n/a           | Minhas 2014      |
| DELPHI                  | 0.55          | 0.33          | n/a           | Li 2021          |
| MaSIF-site              | 0.60          | 0.36          | 0.870         | Gainza 2020      |
| <b>ECABSD V3 (ours)</b> | <b>0.5797</b> | <b>0.5152</b> | <b>0.8928</b> | <b>This work</b> |

## 4.3 Ablation Study

Table 3: Component Ablation (Homology-Filtered Test Set)

| Variant                         | F1            | MCC           | ROC-AUC       | $\Delta$ F1 |
|---------------------------------|---------------|---------------|---------------|-------------|
| <b>Full V3 (all components)</b> | <b>0.5797</b> | <b>0.5152</b> | <b>0.8928</b> | —           |
| No Cross-Attention              | 0.4103        | 0.3541        | 0.7812        | −0.169      |
| GCN instead of GATv2            | 0.4891        | 0.4287        | 0.8341        | −0.091      |
| No Global Pooling               | 0.5412        | 0.4803        | 0.8701        | −0.039      |
| Sequence-only MLP               | 0.3847        | 0.3102        | 0.7405        | −0.195      |

Cross-attention is the most critical component (−0.169 F1). GATv2 outperforms plain GCN (−0.091 F1). Global pooling provides consistent gains (−0.039 F1). Sequence-only MLP is worst, confirming 3D structural context is essential.

## 4.4 5-Fold Cross-Validation

5-fold CV on homology-aware splits: mean F1 =  $0.4673 \pm 0.0077$ , ROC-AUC =  $0.8338 \pm 0.0057$  (20-epoch budget). Low variance confirms training stability. Full 80-epoch training expected to yield F1  $\approx 0.58$ .

## 4.5 Biological Case Study — PDB 1AY7

Applied to 1AY7 (RNase Sa / Barstar, 1.8 Å resolution): 16 residues predicted binding, all 15 true interface residues correctly identified. Precision = 0.938, Recall = 1.000, F1 = 0.968. The single false positive (Arg31, prob=0.541) is 5.1 Å from the nearest Barstar atom — biologically borderline. Predictions clustered into three known interface patches:  $\beta$ -strand loop (37–41), active site loop (64–69), and C-terminal helix contacts (84–87).

## 5. Explainability and Deployment

Grad-CAM Saliency computes gradients of the binding score with respect to input node features, highlighting biochemical regions driving prediction. Attention Rollout extracts cross-fusion weights showing how Chain A

residues attend to Chain B residues. A FastAPI web application (web/app.py) accepts raw .pdb files or RCSB PDB IDs and returns per-residue predictions with heatmaps and PyMOL export scripts.

## 6. Limitations and Future Work

ECABSD has been evaluated primarily on DB5, which is smaller than large-scale interaction datasets. Future work includes:

- Extending training to DIPS and PINDER datasets
- Precision-focused and recall-focused threshold modes
- Top-K precision evaluation and ensemble prediction (V2 + V3)
- Full 80-epoch cross-validation and external blind test set evaluation

## 7. Conclusion

ECABSD presents an interaction-aware framework for per-residue PPI binding site prediction. Integrating ESM-2 embeddings with GATv2 structural representations and partner-aware cross-attention, ECABSD outperforms all published baselines on MCC and ROC-AUC under homology-filtered evaluation. The 1AY7 case study demonstrates near-perfect interface recovery (F1=0.968). Integrated Grad-CAM and Attention Rollout add interpretability supporting downstream biological validation. ECABSD is a promising foundation for virtual screening, protein design, and drug discovery.

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## Declarations

**Data Availability:** Code and data available at <https://github.com/amanigreeva/ECABSD>

**Competing Interests:** The authors declare no competing interests.

**Authors' Contributions:** AM led model development and experiments. DN, KPSR, KVR, and VK contributed to data processing, evaluation, and manuscript preparation. CSB provided mentorship and guidance.